NIRAJ NIRAULA

M.Tech Mechanical Engineering (Design Engineering) — Indian Institute of Technology Bombay

Letang, Morang, Nepal | +977-9825307023 | nirajniraula765@gmail.com | linkedin.com/in/niraj-niraula-IITB | Portfolio

PROFESSIONAL SUMMARY

Mechanical design engineer (M.Tech, IIT Bombay) specializing in robotics and mechanism design, with hands-on experience developing a tendon-driven robotic manipulator for autonomous EV charging — spanning kinematic design, multi-body dynamic simulation (ADAMS), experimental motion-capture validation (Qualisys), and neural-network-based kinematic modeling (PyTorch). Proficient in SolidWorks, ANSYS, and AutoForm across the full design-to-manufacturing workflow, with an undergraduate specialization in Automobile Engineering. Proven ability to design, simulate, fabricate, and validate mechanical systems against quantified engineering targets.

AREAS OF INTEREST

Product Design & Development • Research & Development (R&D) • Finite Element Analysis (FEA) • Mechanism Synthesis & Simulation • Robotics & Multi-Body Dynamics • GD&T • DFMA • FMEA • AI & ML for Engineering

TECHNICAL SKILLS

Programming: Python, MATLAB, Octave

CAD & Design: SolidWorks, Fusion 360, Autodesk Inventor, MasterCam

Simulation & FEA: ANSYS (Structural, CFD), ADAMS (Kinematic/Dynamic), AutoForm, Simulink

Robotics & Controls: Tendon-Driven Actuation, Forward/Inverse Kinematics, Neural Networks (PyTorch), Qualisys Motion Capture

Documentation: LaTeX, MS Office

M.TECH THESIS — FLAGSHIP PROJECT

Robotic Solution for Autonomous EV Charging (TAP) | *Robotics Laboratory, IIT Bombay - Guide: Prof. Anirban Guha*
May 2025 – Jun 2026

- Designed a quaternion-inspired hybrid robotic manipulator with dual-motor tendon-driven actuation for autonomous EV charging-plug alignment, achieving sub-millimeter positional accuracy.
- Architected and compared two manipulator configurations (universal-joint and spherical-joint variants) through systematic workspace analysis to select the optimal design for the target operating volume.
- Built a custom 3D motion-tracking test rig using a Qualisys capture system to record end-effector position and orientation across multiple test configurations.
- Trained neural network models on 10,000+ experimental data points to predict forward/inverse kinematics, achieving validation accuracy within ±1.5 mm RMSE.

CORE ENGINEERING PROJECTS

Design & Manufacturing of Copper Vapor Chamber Thermal Plate | *IIT Bombay- Guide: Prof. Sushil Mishra* Apr 2026

- Designed a copper vapor chamber thermal plate for high-density electronic cooling; completed full CAD modeling in SolidWorks and die/punch tooling design in MasterCam.
- Optimized sheet-metal formability through iterative AutoForm simulations, reducing predicted forming defects and improving manufacturability.

FEA of Honeycomb-Core Structured Cantilever Beam | *IIT Bombay -Guide: Prof. Yogesh M. Desai* Apr 2025

- Performed structural FEA in ANSYS comparing solid vs. honeycomb-core cantilever beam geometries under equivalent loads.
- Quantified a 218.6% improvement in stiffness-to-weight ratio, demonstrating superior material efficiency for lightweight structural design.

CAE Axisymmetric Contact Analysis —Hertzian Theory Validation | *IIT Bombay - Guide: Prof. Yogesh M. Desai* Apr 2025

- Built optimized 2D axisymmetric contact models in SolidWorks and ran non-linear static simulations in ANSYS Workbench.
- Matched simulated peak contact pressure (152.87 MPa) to classical Hertzian theory within 0.05% error through mesh convergence studies.

Design & Fabrication of DRUJ Biomechanical Measurement Jig | *IIT Bombay - Prof. Darshan S. Shah* Apr 2025

- Designed a precision measurement jig for Distal Radioulnar Joint (DRUJ) range-of-motion and torque testing, engineered to tight fixturing tolerances for repeatable gauge accuracy.
- Oversaw fabrication and verified fitment, improving rotational measurement repeatability by 90% over the prior test method.

Kinematic & Dynamic Analysis of Six-Bar Toggle Mechanism | *IIT Bombay - Prof. Anirban Guha* Apr 2025

- Modeled a six-bar toggle mechanism in SolidWorks and ran full kinematic/dynamic simulations in ADAMS, extracting displacement, velocity, acceleration, force, and torque profiles.
- Validated simulation outputs against analytical and graphical solutions with 95% agreement, confirming model fidelity for mechanism synthesis.

Kinematic & Dynamic Analysis of Textile Mechanisms | *IIT Bombay - Prof. Anirban Guha* Mar 2025

- Modeled doffing and beat-up textile mechanisms in SolidWorks and simulated kinematic/dynamic behavior in ADAMS.
- Designed an alternate mechanism topology matching equivalent motion output with 10–20% lower mass and 25–30% reduced peak power consumption.

M.Tech Seminar: Soft Flexible Robotic Manipulators | *IIT Bombay - Prof. Anirban Guha* Jan 2025 Apr 2025

- Synthesized a literature review of 7 peer-reviewed papers covering design, actuation, sensing, and closed-loop control of soft robotic manipulators.
- Proposed an original application framework for autonomous, safe EV charging via a soft robotic arm, forming the conceptual basis for the M.Tech thesis.

NACA-6409 Airfoil Blade Performance on Cross-Flow Turbines | *IOE, Purwanchal - Asst. Prof. Lal Babu Prasad* May 2022 – May 2023

- Designed a cross-flow turbine using NACA-6409 hydrofoil blades in SolidWorks, achieving 72.3% peak hydraulic efficiency — 4% above the standard-blade baseline.
- Ran CFD simulations in ANSYS to optimize chord angle, identifying 29° as the optimal configuration for maximum efficiency.

PROFESSIONAL EXPERIENCE & LEADERSHIP

Engineering Trainee | *Puwakhola Hydropower Plant, Nepal Electricity Authority* May 2023 – Jun 2023

- Inspected turbine cooling systems and documented maintenance procedures to support reliability improvement across generating units.
- Assisted in equipment performance monitoring and preventive maintenance scheduling.

Teaching Assistant — Makerspace Lab (MS101) | *IIT Bombay* Aug 2024 – Dec 2024

- Supervised lab sessions and mentored 2 project groups (12 students) on UAV/drone builds, including structural airframe integrity and balance.
- Evaluated assignments, quizzes, and exams for the course.

Workshop Trainer — SolidWorks & ANSYS | *Diploma Training Program* Aug 2022

- Conducted hands-on training sessions for diploma students covering 3D CAD modeling in SolidWorks and structural simulation in ANSYS.

EDUCATION

M.Tech, Mechanical Engineering — Design Engineering | *Indian Institute of Technology Bombay* 2026

B.Tech, Mechanical Engineering — Automobile Engineering elective | *Tribhuvan University, Purwanchal Campus* 2023

CERTIFICATIONS & LICENSES

- Certified SOLIDWORKS Associate (CSWA), Dassault Systems Link
- Registered Mechanical Engineer, Nepal Engineering Council , cleared the national licensing exam among 10,000+ participants (2023) Link